Regression Problems

Dataset (sales.xls)

A sales manager collected the following data on annual sales for new customer accounts and the number of years of experience for a sample of 10 salespersons.

- Develop a scatter diagram for these data with years of experience as the independent variable.
- Develop an estimated regression equation that can be used to predict annual sales given the years of experience.
- Use the estimated regression equation to predict annual sales for a salesperson with 9 years of experience.

Dataset - GPS

Using a global-positioning-system (GPS)-based navigator for your car, you enter a destination and the system will plot a route, give spoken turn-by-turn directions, and show your progress along the route. Today, even budget units include features previously available only on more expensive models. *Consumer Reports* conducted extensive tests of GPS-based navigators and developed an overall rating based on factors such as ease of use, driver information, display, and battery life. The following data show the price and rating for a sample of 20 GPS units with a 4.3-inch screen that *Consumer Reports* tested (*Consumer Reports* website, April 17, 2012).

- Develop a scatter diagram with price as the independent variable.
- What does the scatter diagram developed in part (a) indicate about the relationship between the two variables?
- Use the least squares method to develop the estimated regression equation.
- Predict the rating for a GPS system with a 4.3-inch screen that has a price of $200.